

Transaction Concept & ACID Properties DBMS

> **Definition:** A **Transaction** is a logical unit of database processing that includes one or more database operations (READ, WRITE). To preserve data integrity, transactions must satisfy all four **ACID Properties**.

1. Detailed Technical Explanation

The ACID Properties:

1. Atomicity

- Enforced by the **Recovery Manager** using Write-Ahead Logging (WAL) and Rollback mechanisms.

2. Consistency

- Enforced by database schema constraints (Primary Key, Foreign Key, `CHECK` constraints) and application logic.

3. Isolation

- Enforced by the **Concurrency Control Manager** using locking or timestamping.

4. Durability

- Enforced by **Log Recovery & Checkpointing** writing updates to non-volatile disk/SSD storage.

2. Transaction State Diagram

3. Core Concepts & Memory Keywords

- **ACID:** Atomicity, Consistency, Isolation, Durability.

- **C**

4. Must-Write Points for Exams

- Atomicity guarantees that partial execution of a transaction is impossible.

- Iso

5. Quick Recall Flow

Active